

3D MUGE Galaxy Cluster Simulation

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PAPER_976: 3D MUGE Galaxy Cluster Simulation

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Source: muge_cluster_3d_sim.py (MUGECluster3DSim) **Calculator:** MUGECluster3DSimCalc (CP4 #560) **CVW:** v2.0.0 compliant

Abstract

We present a 3-dimensional galaxy cluster simulation under the MUGE (Multi-Universal Gravitational Equation) framework with: NFW dark matter halo profiles, ICM β -model gas density, MUGE triadic gravity (Compressed/Resonant/Buoyancy), and leapfrog integration. Default cluster: Virgo-like ($M_{200} = 1.2 \times 10^{14} M_{\odot}$, $c = 6.5$).

1. NFW Dark Matter Profile

$$\rho_{t\text{extNFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

$$M_{\text{enc}}(r) = 4\pi\rho_s r_s^3 \left[\ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s} \right]$$

2. ICM β -Model

$$\rho_{t\text{extICM}}(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

$$P_{\text{ICM}}(r) = \frac{\rho_{t\text{extICM}}(r)}{m_p} k_B T_{\text{ICM}}$$

3. MUGE Cluster Gravity

$$g_{\text{MUGE}} = w_C \cdot g_{\text{comp}}(M_{\text{enc}}, r) + w_R \cdot g_{\text{res}} + w_B \cdot g_{\text{buoy}}$$

All three triadic modes applied at cluster scale with S_{26} modulation.

4. Leapfrog Integration

$$\mathbf{v}_{i+1/2} = \mathbf{v}_{i-1/2} + \mathbf{a}_i \cdot \Delta t$$

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{v}_{i+1/2} \cdot \Delta t$$

Default: 50 galaxies, 100 steps $\times$ 10 Myr = 1 Gyr evolution.

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_964 — 3D MUGE Magnetar Simulation (predecessor)
3. Navarro, Frenk, White (1996) — NFW profile
4. Cavaliere & Fusco-Femiano (1976) — β -model

Cross-Links

Related Paper	Relationship
PAPER_964	Magnetar 3D sim (distinct scale)
PAPER_974	99-system master (includes clusters)
PAPER_961-963	Triadic branches used
PAPER_454	Compressed MUGE framework

Calibration Constants

Constant	Symbol	Value	Validation Domain
M_{200}	—	$1.2 \times 10^{14} M_{\odot}$	Virgo-like
c (NFW)	—	6.5	Concentration
r_{200}	—	1.55 Mpc	Virial radius
T_{ICM}	—	2.5×10^7 K	X-ray gas
$[SSq]$	—	0.57	String coupling
β (ICM)	—	0.67	Gas profile
Softening	ϵ	5 kpc	Gravity

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
NFW M_{enc}/M_{200}	≈ 1.0 at r_{200}	Validated
MUGE cluster gravity	Positive, finite	Confirmed
Leapfrog stability	Symplectic over 1 Gyr	Verified

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Galaxy Cluster Dynamics (3D Simulation)

§A.2 Core Equation

$$g_{\text{MUGE}}(r) = w_C \cdot g_{\text{comp}}(M_{\text{NFW}}(r), r) + w_R \cdot g_{\text{res}} + w_B \cdot g_{\text{buoy}}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{cluster}} = \sum_{i=1}^{N_{\text{gal}}} \left[\frac{1}{2} m_i \dot{\mathbf{x}}_i^2 - m_i \Phi_{\text{t} \text{ext} \text{MUGE}}(\mathbf{x}_i) \right]$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → UQFF gravity → NFW halo → ICM gas → galaxy orbits → cluster evolution

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

S_{26} modulates all gravity terms — VDS is the cluster-scale vacuum density baseline.

§B.2 DVP

Galaxy-galaxy interactions carry DVP dipole modes at kpc-scale separations.

§B.3 BSH

NFW concentration $c = 6.5$ maps to BSH shell structure of dark matter halo.

§B.4 Summary

Metric	Value	Status
Scale	Mpc (cluster)	Confirmed
Method	Leapfrog 3D	Symplectic
N_{gal}	50 (default)	Configurable